

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

Student Name: Sautrik Roy	Reg No: RA2511003010052
Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit III | Session 1 | SLO 1: Introduction to Concurrent Programming (Code: 3011)

Q: 1. Fill in the blank:

In concurrent programming, multiple _____ can be executed in overlapping time periods.

Ans: Answer: threads (or processes / tasks)

Q: 2. True or False:

A thread is a lightweight process used in concurrent programming.

Ans: Answer: True

Q: 3. Multiple Choice:

Which of the following is commonly used for managing concurrency?

A) Semaphore B) Compiler C) Keyboard D) Monitor

Ans: Answer: A) Semaphore (and D is also a concurrency construct; Semaphore is the standard synchronization primitive).

Q: 4. Fill in the blank:

A _____ condition occurs when two or more threads wait for each other indefinitely.

Ans: Answer: deadlock

Q: 5. True or False:

Concurrency and parallelism refer to the exact same concept.

Ans: Answer: False (Concurrency is dealing with lots of things at once; parallelism is doing lots of things at once simultaneously).

Q: 6. Fill in the blank:

A _____ occurs when two threads simultaneously modify shared data leading to inconsistent states.

Ans: Answer: race condition

Q: 7. True or False:

Threads belonging to the same process share the same memory space.

Ans: Answer: True

Q: 8. Fill in the blank:

Ensuring that multiple threads execute coordinated access to shared variables is called thread _____.

Ans: Answer: synchronization

Q: 9. Multiple Choice:

A mutual exclusion mechanism that allows only one thread into a critical section is called:

A) Mutex B) JVM C) Garbage Collector D) Compiler

Ans: Answer: A) Mutex

Q: 10. Fill in the blank:

A _____ ensures that only one thread accesses a critical section at a time.

Ans: Answer: mutex (or lock / monitor)

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Unit III | Session 1 | SLO 2: Thread Lifecycle and States (Code: 3012)

Q: 1. Fill in the blank:

A thread in Java is created in the _____ state before start() is invoked.

Ans: Answer: NEW

Q: 2. True or False:

Calling run() directly creates a new operating system thread.

Ans: Answer: False (Calling run() executes the method in the caller's thread; start() creates a new thread).

Q: 3. Multiple Choice:

Which method must be called to begin thread execution?

A) run() B) start() C) init() D) execute()

Ans: Answer: B) start()

Q: 4. Fill in the blank:

When a thread is waiting to acquire a monitor lock, it is in the _____ state.

Ans: Answer: BLOCKED

Q: 5. Programming – Output:

```
Thread t = new Thread(() -> System.out.println("Thread running"));  
t.start();
```

Ans: Output: Thread running

Q: 6. True or False:

A terminated thread can be restarted by calling start() again.

Ans: Answer: False (Throws IllegalStateException).

Q: 7. Multiple Choice:

When the run() method finishes executing, the thread transitions to which state?

A) WAITING B) RUNNABLE C) TERMINATED D) BLOCKED

Ans: Answer: C) TERMINATED

Q: 8. Fill in the blank:

The method Thread._____(ms) pauses execution for specified milliseconds.

Ans: Answer: sleep

Q: 9. Identify Error:

```
Thread t = new Thread(); t.start(); t.start();
```

Ans: Answer: Throws IllegalStateException at runtime.

Q: 10. Fill in the blank:

The exact scheduling order of multiple runnable threads is _____.

Ans: Answer: non-deterministic (or determined by OS thread scheduler)

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Unit III | Session 2 | SLO 1: Thread Synchronization & Locks (Code: 3021)

Q: 1. Fill in the blank:

The _____ keyword in Java prevents concurrent access to a shared method or block.

Ans: Answer: synchronized

Q: 2. True or False:

Every Java object has an associated intrinsic lock (monitor).

Ans: Answer: True

Q: 3. Multiple Choice:

Which of the following guarantees atomic compound operations?

A) AtomicInteger B) int C) double D) boolean

Ans: Answer: A) AtomicInteger

Q: 4. Fill in the blank:

A thread acquires an _____ lock when entering a synchronized block.

Ans: Answer: intrinsic (or monitor)

Q: 5. What problem occurs if two threads increment an unsynchronized shared counter simultaneously?

Ans: Answer: Race Condition (Lost Update anomaly).

Q: 6. True or False:

Static synchronized methods lock on the Class object.

Ans: Answer: True

Q: 7. Fill in the blank:

The _____ class in `java.util.concurrent.locks` provides explicit locking mechanisms.

Ans: Answer: ReentrantLock

Q: 8. Fill in the blank:

The _____ keyword ensures visibility of changes to variables across threads.

Ans: Answer: volatile

Q: 9. Identify the issue with: `synchronized(new Object()) { ... }`

Ans: Answer: Ineffective locking because every thread synchronizes on a brand new object instance.

Q: 10. Fill in the blank:

ReentrantLock allows a thread to re-acquire a lock it already holds without causing _____.

Ans: Answer: deadlock

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Unit III | Session 2 | SLO 2: Inter-Thread Communication (Code: 3022)

Q: 1. Fill in the blank:

The wait(), notify(), and notifyAll() methods belong to the _____ class.

Ans: Answer: Object

Q: 2. True or False:

wait() releases the monitor lock while the thread is waiting.

Ans: Answer: True

Q: 3. Multiple Choice:

Which method wakes up all threads waiting on an object's monitor?

A) notify() B) resume() C) notifyAll() D) wakeAll()

Ans: Answer: C) notifyAll()

Q: 4. Fill in the blank:

wait() must always be called inside a _____ context.

Ans: Answer: synchronized

Q: 5. What exception is thrown if wait() is called without holding the monitor lock?

Ans: Answer: IllegalMonitorStateException

Q: 6. True or False:

Thread.sleep() releases locks while sleeping.

Ans: Answer: False (sleep() holds locks, whereas wait() releases the lock).

Q: 7. Fill in the blank:

To guard against spurious wakeups, wait() should always be placed inside a _____ loop.

Ans: Answer: while

Q: 8. Multiple Choice:

Which classical synchronization problem demonstrates inter-thread communication?

A) Producer-Consumer B) QuickSort C) Binary Search D) Bubble Sort

Ans: Answer: A) Producer-Consumer

Q: 9. Identify the bug: Calling notify() outside synchronized block.

Ans: Answer: Results in IllegalMonitorStateException at runtime.

Q: 10. Fill in the blank:

notify() wakes up an _____ thread waiting on the object monitor.

Ans: Answer: arbitrarily chosen (single)

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Unit III | Session 3 | SLO 1: Thread Pooling and Executors (Code: 3031)

Q: 1. Fill in the blank:

The _____ interface represents an asynchronous execution service in `java.util.concurrent`.

Ans: Answer: `ExecutorService`

Q: 2. True or False:

Creating new threads on demand for every task introduces significant performance overhead.

Ans: Answer: True

Q: 3. Multiple Choice:

Which factory method creates a pool with a set number of threads?

A) `newCachedThreadPool()` B) `newFixedThreadPool()` C) `newSingleThreadExecutor()` D) `newScheduledThreadPool()`

Ans: Answer: B) `newFixedThreadPool()`

Q: 4. Fill in the blank:

Reusing worker threads reduces the overhead of thread _____ and destruction.

Ans: Answer: creation (allocation)

Q: 5. Programming – What happens if `executor.shutdown()` is not called?

Ans: Answer: The JVM may not terminate because active non-daemon pool threads prevent shutdown.

Q: 6. True or False:

`ExecutorService.shutdownNow()` attempts to stop all actively executing tasks.

Ans: Answer: True

Q: 7. Fill in the blank:

The core implementation class behind Executors factory methods is _____.

Ans: Answer: `ThreadPoolExecutor`

Q: 8. Multiple Choice:

Which method is used to submit a task to an `ExecutorService`?

A) `run()` B) `execute()` / `submit()` C) `start()` D) `begin()`

Ans: Answer: B) `execute()` / `submit()`

Q: 9. Fill in the blank:

A pool queue that holds tasks waiting for available threads is typically a _____.

Ans: Answer: `BlockingQueue`

Q: 10. Fill in the blank:

Thread pools prevent resource exhaustion by limiting the maximum number of _____ threads.

Ans: Answer: concurrent (active)

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Unit III | Session 3 | SLO 2: Callable and Future (Code: 3032)

Q: 1. Fill in the blank:

Unlike Runnable, the _____ interface can return a computed result.

Ans: Answer: Callable (java.util.concurrent.Callable)

Q: 2. True or False:

Callable's call() method can throw checked exceptions.

Ans: Answer: True

Q: 3. Multiple Choice:

What method does the Callable interface define?

A) run() B) execute() C) call() D) compute()

Ans: Answer: C) call()

Q: 4. Fill in the blank:

A _____ object represents the pending result of an asynchronous computation.

Ans: Answer: Future

Q: 5. What does the Future.get() method do if the task is still calculating?

Ans: Answer: It blocks the calling thread until computation finishes.

Q: 6. True or False:

Future.isDone() returns true if the task completed normally, threw an exception, or was cancelled.

Ans: Answer: True

Q: 7. Multiple Choice:

Which method cancels task execution?

A) stop() B) abort() C) cancel() D) terminate()

Ans: Answer: C) cancel()

Q: 8. Fill in the blank:

Future.get(timeout, unit) throws _____ if the timeout expires before completion.

Ans: Answer: TimeoutException

Q: 9. Identify Error: Callable c = () -> { return 5; }; Thread t = new Thread(c);

Ans: Answer: Thread constructor accepts Runnable, not Callable directly (use FutureTask or ExecutorService).

Q: 10. Fill in the blank:

Callable is a parameterized functional interface with generic type _____.

Ans: Answer: Callable<V> (or <T>)

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Unit III | Session 4 | SLO 1: Thread Priorities and Daemon Threads (Code: 3041)

Q: 1. Fill in the blank:

The default priority of a thread in Java is _____.

Ans: Answer: 5 (Thread.NORM_PRIORITY)

Q: 2. True or False:

Daemon threads continue running even after all user threads have finished execution.

Ans: Answer: False (JVM terminates automatically when all user threads finish).

Q: 3. Multiple Choice:

Which method is used to set the priority of a thread?

A) setOrder() B) setLevel() C) setPriority() D) setThreadLevel()

Ans: Answer: C) setPriority()

Q: 4. Fill in the blank:

The priority range for threads in Java is from _____ to _____.

Ans: Answer: 1 (MIN_PRIORITY) to 10 (MAX_PRIORITY)

Q: 5. Programming – Find output:

```
Thread t = new Thread(() -> System.out.println(Thread.currentThread().isDaemon()));  
t.setDaemon(true); t.start();
```

Ans: Output: A) true

Q: 6. True or False:

Calling setDaemon(true) on a thread after start() has been called will throw an exception.

Ans: Answer: True (IllegalThreadStateException)

Q: 7. Multiple Choice:

Which thread type is typically used for background services like garbage collection?

A) Foreground B) User thread C) Daemon thread D) High-priority

Ans: Answer: C) Daemon thread

Q: 8. Fill in the blank:

The method _____ returns whether a thread is a daemon.

Ans: Answer: isDaemon()

Q: 9. Identify Error:

```
Thread t = new Thread(() -> System.out.println("Running")); t.start(); t.setDaemon(true);
```

Ans: Answer: C) IllegalStateException (setDaemon must precede start()).

Q: 10. Fill in the blank:

Threads with higher priority are _____ likely to be scheduled before lower priority threads.

Ans: Answer: more

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Unit III | Session 4 | SLO 2: Runnable Interface (Code: 3042)

Q: 1. Fill in the blank:

In Java, a class implements the _____ interface to create a thread using Runnable.

Ans: Answer: Runnable

Q: 2. True or False:

The run() method must be defined when implementing the Runnable interface.

Ans: Answer: True

Q: 3. Multiple Choice:

Which is the correct way to create a thread using Runnable?

A) new Thread(new MyRunnable()).start(); B) new Runnable().start(); C) new Thread().run(); D) new MyRunnable().start();

Ans: Answer: A) new Thread(new MyRunnable()).start();

Q: 4. Fill in the blank:

The Runnable interface belongs to the java.lang _____.

Ans: Answer: package

Q: 5. True or False:

The Runnable interface can be used to create multiple threads sharing the same task.

Ans: Answer: True

Q: 6. Multiple Choice:

What keyword is used when a class uses the Runnable interface?

A) extends B) implements C) imports D) inherits

Ans: Answer: B) implements

Q: 7. Fill in the blank:

The run() method of a Runnable object is invoked by the _____ class.

Ans: Answer: Thread

Q: 8. True or False:

Runnable allows better separation of thread creation and task definition.

Ans: Answer: True

Q: 9. Multiple Choice:

Which of the following is an advantage of using Runnable over extending Thread?

A) Allows multiple inheritance B) Allows sharing the same task between threads C) Direct JVM access D) Overriding main

Ans: Answer: B) Allows sharing the same task between threads (and A is also a primary benefit).

Q: 10. Fill in the blank:

A thread is started by calling the _____ method on the Thread object.

Ans: Answer: start()

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Unit III | Session 5 | SLO 1: Declarative Programming in Databases (Code: 3051)

Q: 1. Fill in the blank:

In declarative programming, the focus is on _____ rather than how the task is performed.

Ans: Answer: what (the desired result)

Q: 2. True or False:

SQL is an example of declarative programming language used in databases.

Ans: Answer: True

Q: 3. Multiple Choice:

Which of the following best represents a declarative SQL statement?

A) Use for-loop to iterate B) SELECT name FROM students WHERE grade > 90; C) Stored procedure loop D) Cursor reading

Ans: Answer: B) SELECT name FROM students WHERE grade > 90;

Q: 4. Fill in the blank:

Declarative programming abstracts away the _____ of data access.

Ans: Answer: internal implementation details (procedural algorithms)

Q: 5. SQL Task – Find Output:

Table: Students (1 Alice 85, 2 Bob 92, 3 Clara 88)

Query: SELECT name FROM Students WHERE grade > 90;

Ans: Output: B) Bob

Q: 6. True or False:

In declarative programming, the database engine determines the most efficient way to execute a query.

Ans: Answer: True (via the Query Optimizer and Execution Plan).

Q: 7. Multiple Choice:

Which of these is NOT a feature of declarative programming in databases?

A) Focus on result B) Automatic optimization C) Explicit control flow D) Use of queries

Ans: Answer: C) Explicit control flow

Q: 8. Fill in the blank:

The WHERE clause in SQL is used to _____ records in declarative queries.

Ans: Answer: filter

Q: 9. SQL Task – Identify Error:

SELECT * FROM students WHERE grade => 90;

Ans: Answer: C) Invalid operator (syntax error: operator must be >= not =>).

Q: 10. Fill in the blank:

Declarative programming helps developers express logic without describing its _____.

Ans: Answer: execution control flow (steps)

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Unit III | Session 5 | SLO 2: Architecture of JDBC in Java (Code: 3052)

Q: 1. Fill in the blank:

The JDBC API provides a _____-independent way to access databases from Java.

Ans: Answer: database (or vendor)

Q: 2. True or False:

The DriverManager class is responsible for managing a list of database drivers.

Ans: Answer: True

Q: 3. Multiple Choice:

Which component acts as the interface between a Java application and the database driver?

A) Connection B) ResultSet C) Statement D) DriverManager

Ans: Answer: D) DriverManager

Q: 4. Fill in the blank:

The method used to register a JDBC driver class manually is Class.forName("_____").

Ans: Answer: driver class name (e.g., com.mysql.cj.jdbc.Driver)

Q: 5. Programming – Output:

```
Connection con = DriverManager.getConnection(url, user, pass);
```

```
if (con != null) System.out.println("Connected");
```

Ans: Output: Connected (assuming valid database connection).

Q: 6. True or False:

The ResultSet object allows forward and backward navigation by default.

Ans: Answer: False (Default ResultSet is forward-only).

Q: 7. Multiple Choice:

Which interface is used to execute static SQL queries?

A) Statement B) CallableStatement C) PreparedStatement D) Connection

Ans: Answer: A) Statement

Q: 8. Fill in the blank:

The method executeQuery() of Statement returns a _____ object.

Ans: Answer: ResultSet

Q: 9. Programming – Identify Error:

```
Statement stmt = con.createStatement();
```

```
ResultSet rs = stmt.executeUpdate("SELECT * FROM users");
```

Ans: Answer: B) executeQuery should be used (executeUpdate returns an int, not ResultSet).

Q: 10. Fill in the blank:

The JDBC architecture consists of two main layers: the JDBC API and the JDBC _____.

Ans: Answer: Driver API